STAFF REVIEW · H446 114 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. No. Price is not the same as licence/source rights.

2. Sum 0 and carry-out 1.

3. 10.

4. A link/reference to the next node (for a singly linked list).

I Do — source worked steps

1. Model answer — A compiler translates source to target machine/object code; linked executable can run without the source being present. Compilation can report syntax errors and apply optimisation before execution. — Check: link each claim to the scenario and use precise OCR terminology.

We Do — guided CHECK

1. What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.
Work through the method
Use the middle stage of the visual to follow the process and state each intermediate result.
Shared worked answer
An assembler converts the mnemonic plus operand into its encoded machine instruction. In standard LMC, LDA 20 encodes as 520. This is not an interpreter translating a high-level language.

You Do — source mark schemes

1. Compiler high-level source to target code [2], interpreter execution-time evaluation [2], assembler assembly mnemonics to machine/object code [2].

2. Potential runtime performance [2], distribution without original source [2], early compile-time error detection/optimisation [2].

3. Correct translator characteristics [3], specific debugging trade-off [2], portability/target dependence [2], qualified conclusion [1].

Exit Ticket — source answer

1. False. Assemblers translate assembly language.